

Why $\hat{\theta}_n$ Actually Works: The LLN

Theorem (Strong Law of Large Numbers)

If $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} f$ and $E_f|h(X)| < \infty$, then

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i) \xrightarrow{\text{a.s.}} \theta = E_f[h(X)] \quad \text{as } n \rightarrow \infty.$$

- This is what justifies calling $\hat{\theta}_n$ a **consistent** estimator: more simulations $\Rightarrow$ the answer gets arbitrarily close to the truth, with probability 1.
- Unbiasedness (every finite n) and consistency (as $n \rightarrow \infty$) together are the two pillars that make crude Monte Carlo trustworthy.
- The LLN says *nothing* about the finite-sample error, only that it eventually vanishes – for that we need the CLT.

Quantifying Stochastic Error

Suppose we estimate $\theta = E_f[h(X)]$ by the sample mean

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- **Central Limit Theorem:** if $\sigma^2 = \text{Var}_f[h(X)] < \infty$, then for large n ,

$$\hat{\theta}_n \stackrel{\text{approx}}{\sim} N\left(\theta, \frac{\sigma^2}{n}\right).$$

- **Monte Carlo Standard Error:**

$$\text{SE}(\hat{\theta}_n) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}},$$

where s is the sample standard deviation of $h(X_1), \dots, h(X_n)$.

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Approximate $(1 - \alpha)$ confidence interval

$$\hat{\theta}_n \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}.$$

The $1/\sqrt{n}$ Rate – Blessing and Curse

Key fact

Monte Carlo error shrinks at rate $O(1/\sqrt{n})$, **regardless of the dimension** of X .

Recall: $\theta = E_f(h(x)) = \int h(x) f(x) dx \rightarrow$

$$\Rightarrow \int g(x) dx \Rightarrow \int \frac{g(x)}{f(x)} f(x) dx$$

$$\Rightarrow E_f\left[\frac{g(x)}{f(x)}\right]$$

crude monte carlo $\leftarrow$

$$X_1, \dots, X_n \stackrel{iid}{\sim} f$$

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i) \rightarrow \text{crude mc estimator}$$

$\hookrightarrow$ to estimate θ

Properties

(i) unbiasedness $\rightarrow E(\hat{\theta}_n) = \theta = \int h(x) f(x) dx = E_f(h(x))$

(ii) consistency $\rightarrow \hat{\theta}_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$

$\hookrightarrow$ $\frac{SLLN}{\text{as}} \Rightarrow \omega LLN \xrightarrow{P}$

$$\hat{\theta}_n \xrightarrow{P} \theta \text{ as } n \rightarrow \infty \text{ (SLLN)}$$

$$\Rightarrow \hat{\theta}_n \xrightarrow{P} \theta \text{ (WLLN)} \Rightarrow \text{Consistency}$$

$\hookrightarrow$ classical WLLN $\rightarrow$

$$X_1, \dots, X_n \stackrel{iid}{\sim} F, E(X_i) = \mu, \text{ s.t. } \underline{Var}(X_i) = \sigma^2 < \infty$$

$$\Rightarrow \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} E(X) = \mu$$

$\hookrightarrow$ for CMC $\rightarrow \hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i)$

since $X_1, \dots, X_n \stackrel{iid}{\sim} f \Rightarrow h(X_1), \dots, h(X_n) \stackrel{iid}{\sim}$

Assume $\left\{ \begin{array}{l} E_f(h(X_i)) = \mu = \int h(x) f(x) dx \\ Var_f[h(X_i)] = \sigma^2 < \infty \end{array} \right.$

$$\Rightarrow \hat{\theta}_n \xrightarrow{P} \theta$$

$\Downarrow$

$$\sigma^2 = E(h(X_i)^2) - (E(h(X_i)))^2$$

$\hookrightarrow$ assume

(iii) $\Rightarrow$ CLT $\Rightarrow$

classical CLT $\rightarrow X_1 \dots X_n \stackrel{iid}{\sim} F$ $E(X_i) = \mu, V(X_i) = \sigma^2 < \infty$
 $\Rightarrow Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \rightarrow N(0, 1)$ as $n \rightarrow \infty$

$\Rightarrow$ crude monte carlo [cmc]

$\Rightarrow \hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i) \rightarrow \theta = E_f[h(X)] = \int h(x)f(x)dx$

$\Rightarrow X_1 \dots X_n \stackrel{iid}{\sim} F \Rightarrow$
 Assumption $\begin{cases} E_f[h(X_i)] = \theta \\ Var_f[h(X_i)] = \sigma^2 < \infty \end{cases}$
 $\int h(x_1)f(x_1)dx_1 = \int h(x_2)f(x_2)dx_2 = \dots = \int h(x_n)f(x_n)dx_n$

$X_1 \dots X_n \stackrel{iid}{\sim} F \Rightarrow h(X_1) \dots h(X_n) \stackrel{iid}{\sim}$

$\Rightarrow Z_n = \frac{\hat{\theta}_n - \theta}{\sigma/\sqrt{n}} \rightarrow N[0, 1], \text{ as } n \rightarrow \infty$

$\Rightarrow \hat{\theta}_n \sim N\left[\theta, \frac{\sigma^2}{n}\right], \sigma^2 = Var_f[h(X_i)]_{i=1:n}$

$\hat{\theta}_n = Z_n \frac{\sigma}{\sqrt{n}} + \theta$

$E(\hat{\theta}_n) = \theta$
 $\hat{\sigma}^2 = Var(Z_n \frac{\sigma}{\sqrt{n}})$
 $Var(\hat{\theta}_n) = \frac{\sigma^2}{n} Var(Z_n) = \frac{\sigma^2}{n}$

(iv) $\rightarrow SE(\hat{\theta}_n) = \sqrt{Var(\hat{\theta}_n)} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$

sample se. $\rightarrow \hat{\sigma} = s \rightarrow$ sqrt of $= \sqrt{\frac{1}{n-1} \sum_{i=1}^n (h(X_i) - \hat{\theta}_n)^2}$ Sample Variance

(v) $\Rightarrow$ confidence interval $\Rightarrow$ 95% CI

$\hat{\theta}_n \pm Z_{\alpha/2} SE(\hat{\theta}_n)$

$= \hat{\theta}_n \pm 1.96 \times \frac{s}{\sqrt{n}}$

$\Rightarrow (\hat{\theta}_n - 1.96 \frac{s}{\sqrt{n}}, \hat{\theta}_n + 1.96 \frac{s}{\sqrt{n}})$

The $1/\sqrt{n}$ Rate – Blessing and Curse

Key fact

Monte Carlo error shrinks at rate $O(1/\sqrt{n})$, **regardless of the dimension** of X .

- **Blessing:** unlike classical quadrature (whose cost explodes with dimension – the “curse of dimensionality” from the previous section), Monte Carlo’s convergence rate is *dimension-free*. This is *why* simulation is indispensable in the rendering, finance, and Bayesian examples we opened with.

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- **Curse:** to halve the standard error, you must **quadruple** the number of simulations ($n \rightarrow 4n$ gives $SE \rightarrow SE/2$) – convergence is comparatively slow, and each extra digit of accuracy is expensive.

$$\Rightarrow SE(\hat{\theta}_n) = \sigma/\sqrt{n} \rightarrow S/\sqrt{n}$$

half $\rightarrow$ $\frac{S}{2\sqrt{n}} \rightarrow \frac{S}{\sqrt{4n}}$
 $n^* = 4n$

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The lever we actually control

Since $SE = \sigma/\sqrt{n}$, there are only two ways to shrink it: increase n (expensive), or **reduce** σ^2 by sampling more cleverly. The rest of this module is devoted to the second option – importance sampling and variance reduction.



Reporting Simulation Results Honestly

Good practice

Never report a Monte Carlo estimate as a bare number. Always accompany it with a measure of its uncertainty:

$$\hat{\theta}_n \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \quad (\text{an approximate } (1 - \alpha) \text{ confidence interval}).$$

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- Increasing n should shrink the interval at the theoretical $1/\sqrt{n}$ rate; if it does not, suspect a bug or an $h(X)$ with infinite variance.
- We return to this with concrete diagnostics at the end of the module.

Outline

Motivation: The Integrals We Cannot Solve

The Monte Carlo Principle

Law of Large Numbers and Monte Carlo Error

Crude Monte Carlo in Practice

Estimating Probabilities

The Rare-Event Problem

Importance Sampling

Variance Reduction Ideas

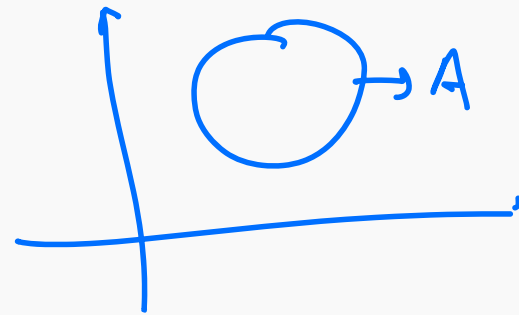
Practical Monte Carlo Diagnostics

Unit Summary

Special Case: Estimating a Probability

- Suppose $X \sim f$ and we want the probability that X lands in some event/region A .

$$P(X \in A)$$
$$X = (X_1, X_2)$$
$$P(X \in A)$$



Special Case: Estimating a Probability

- Suppose $X \sim f$ and we want the probability that X lands in some event/region A .
- We will simulate $X_1, \dots, X_n$ **i.i.d. from** f (i.e., from the distribution that defines θ).

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- Take $h(x) = \mathbf{1}\{x \in A\}$. Then

$$\theta = E_f[\mathbf{1}\{X \in A\}] = P(X \in A).$$

$$h(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{o.w.} \end{cases}$$

$$E_f(h(X)) = 1 \cdot P(X \in A) + 0 \cdot \dots = P(X \in A)$$

Alg $\rightarrow$ step 1 $\rightarrow X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} f$

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \in A\}$$

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- Crude Monte Carlo estimator: $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \in A\}$ – simply the **sample proportion** of draws landing in A .

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- Since $\mathbf{1}\{X \in A\}$ is Bernoulli(θ), its variance is $\theta(1 - \theta)$, so

$$SE(\hat{\theta}_n) = \sqrt{\frac{\theta(1 - \theta)}{n}}.$$

$\int_0^1$
 $\downarrow$

$$V(\hat{\theta}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \in A\}\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(\mathbf{1}\{X_i \in A\})$$

$$= \frac{1}{n^2} \sum_{i=1}^n \theta(1 - \theta) = \frac{\theta(1 - \theta)}{n}$$

$$\begin{aligned} \theta &\rightarrow P(X \in A) \\ &= E_f[h(X)] \\ &= \int \mathbf{1}\{x \in A\} f(x) dx \end{aligned}$$

When Crude Monte Carlo Struggles: Rare Events

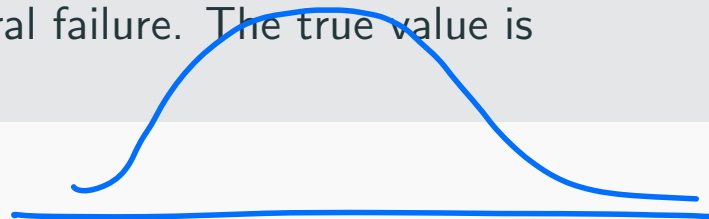
Motivating example

A structural engineer needs $\theta = P(X > 6)$ where $X \sim N(0, 1)$ represents a (standardized) stress level, and $\{X > 6\}$ is catastrophic structural failure. The true value is $\theta \approx 9.87 \times 10^{-10}$.

$$A = (6, \infty)$$



$$P(X \in A) \approx \frac{1}{n} \cdot \sum_{i=1}^n I(X_i \in A)$$



$$P(-3 < X < 3) = 0.99$$

$$P(X > 3) \approx 0.01$$

$$P(X > 4) \approx 10^{-5}$$

$$P(X > 6) \approx 10^{-10}$$

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- Crude MC: draw $X_i \sim N(0, 1)$, estimate $\hat{\theta}_n = \frac{1}{n} \sum \mathbf{1}\{X_i > 6\}$. ✓
- Relative standard error $\frac{\text{SE}(\hat{\theta}_n)}{\theta} \approx \frac{1}{\sqrt{n\theta}}$ – to get even a modestly reliable relative error, we would need n in the **billions**. ✓
- In practice: with $n = 10^6$ draws, we expect to see zero exceedances at all – the estimator returns $\hat{\theta}_n = 0$, which is useless.

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- In practice: with $n = 10^6$ draws, we expect to see *zero* exceedances at all – the estimator returns $\hat{\theta}_n = 0$, which is useless.

The lesson

Crude Monte Carlo wastes almost all its draws on the “uninteresting” region and almost never visits the region that matters. This is precisely analogous to the naive-rejection failure we saw for truncated Normal sampling in Module 1 – and it motivates the next section.

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Importance Sampling

Motivation and the Change-of-Measure Idea

Simple (Unbiased) Importance Sampling

Variance and the Optimal Proposal

Worked Example: Rescuing the Rare-Event Problem

Weighted (Self-Normalized) Importance Sampling

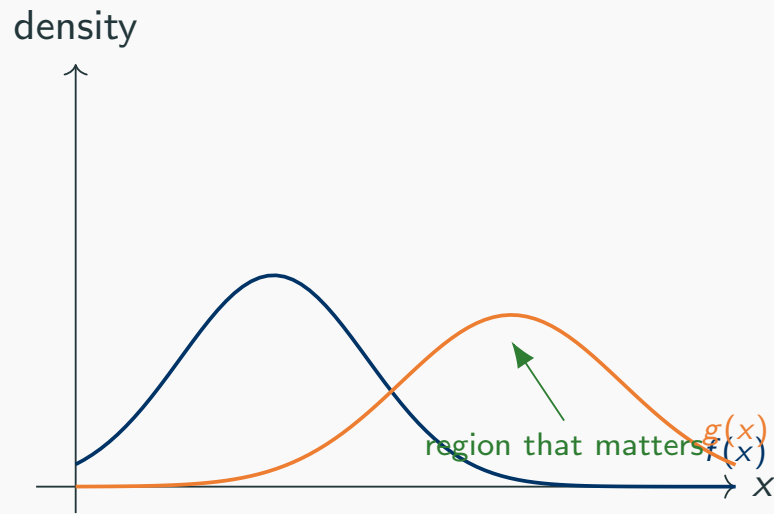
Variance Reduction Ideas

Practical Monte Carlo Diagnostics

The Idea: Sample Where It Matters

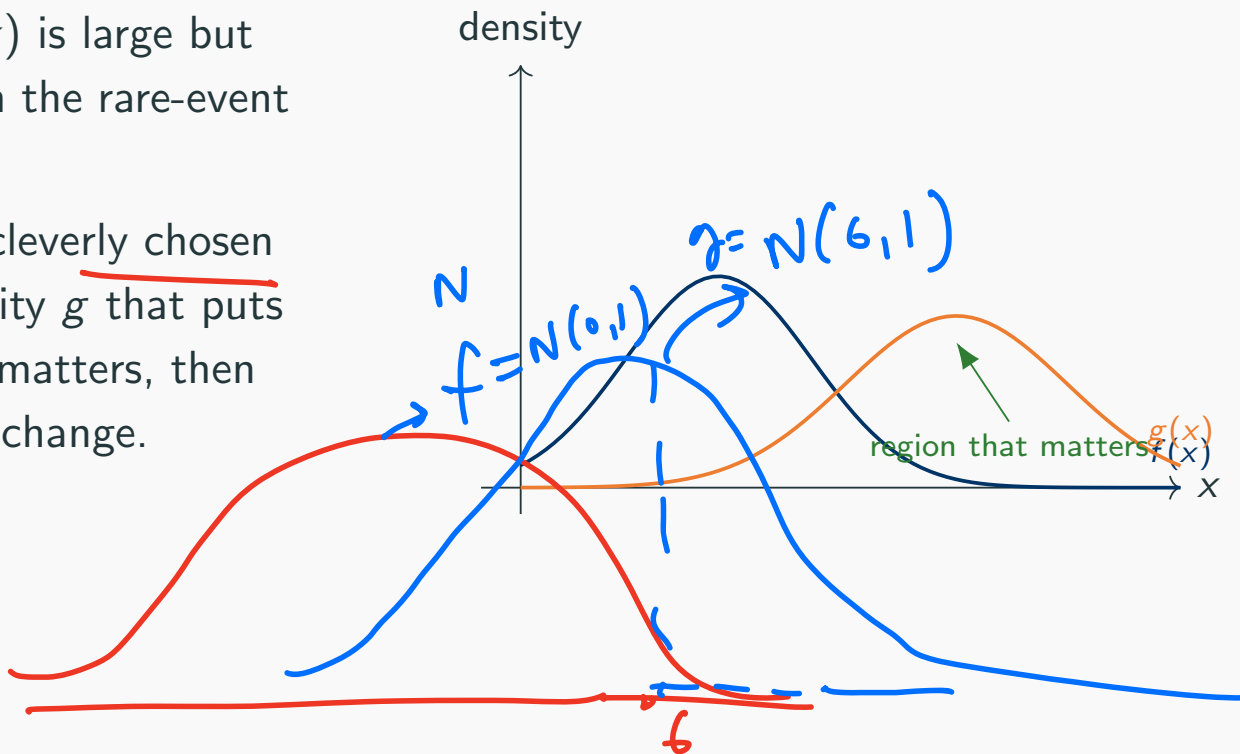
$$\theta = \int h(x) f(x) dx = E f(h(x))$$

- Crude MC draws from f itself, so it rarely visits regions where $h(x)f(x)$ is large but $f(x)$ is small (e.g. the tail in the rare-event example).



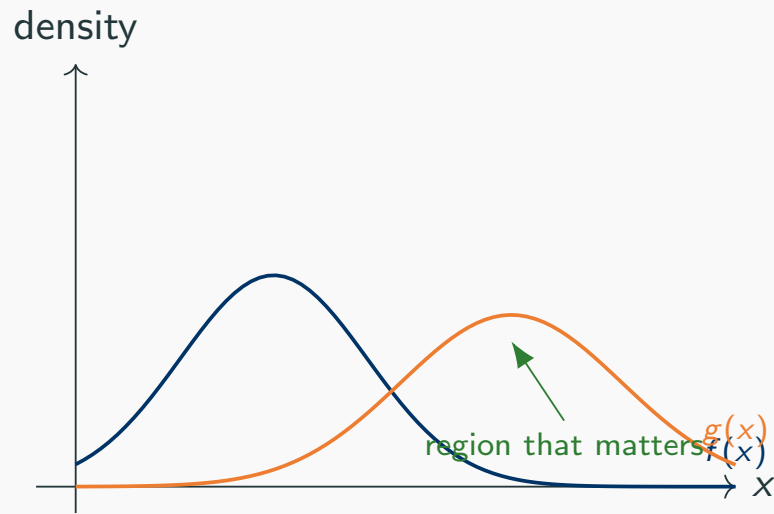
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- **Idea:** draw instead from a cleverly chosen importance/proposal density g that puts more mass where $h(x)f(x)$ matters, then **reweight** to correct for the change.



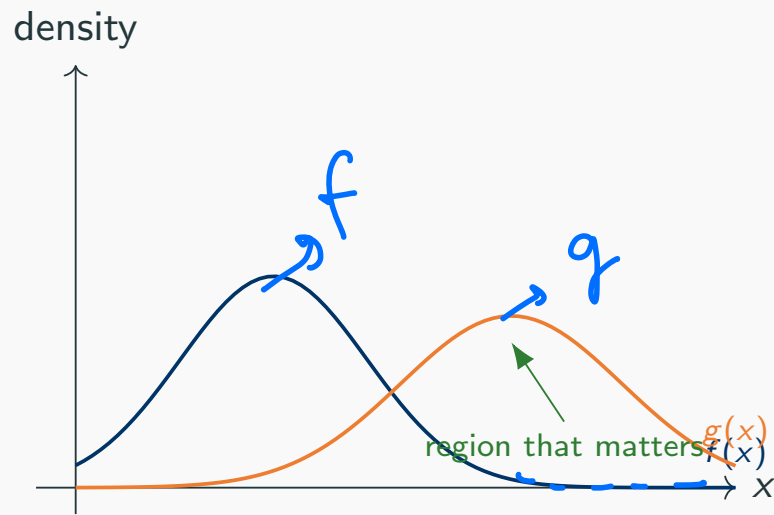
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- **Idea:** draw instead from a cleverly chosen **importance/proposal** density g that puts more mass where $h(x)f(x)$ matters, then **reweight** to correct for the change.



This exactly mirrors the acceptance–rejection idea from Module 1 (choose a helpful “envelope”), except now we *keep every draw* and correct with a weight, rather than throwing draws away.

The Importance Sampling Identity

Theorem (Importance Sampling)

Let g be any density with $g(x) > 0$ whenever $h(x)f(x) \neq 0$. Then

$$\theta = E_f[h(X)] = \int h(x)f(x) dx = \int h(x) \frac{f(x)}{g(x)} g(x) dx = E_g \left[h(X) \frac{f(X)}{g(X)} \right].$$

$\Downarrow$

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$\Rightarrow$ Stable $x_1 \dots x_n \stackrel{\text{iid}}{\sim} g$ $\rightarrow$ weight

$$\Rightarrow \theta_n = \frac{1}{n} \sum_{i=1}^n \frac{h(x_i) f(x_i)}{g(x_i)} = \frac{1}{n} \sum_{i=1}^n w_i h(x_i), \quad w_i = \frac{f(x_i)}{g(x_i)}$$

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Algorithm: Simple Importance Sampling

1. Generate $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} g$.

2. Compute **importance weights** $w_i = \frac{f(X_i)}{g(X_i)}$. $\rightarrow$

3. Return

$$\hat{\theta}_n^{\text{IS}} = \frac{1}{n} \sum_{i=1}^n w_i h(X_i).$$

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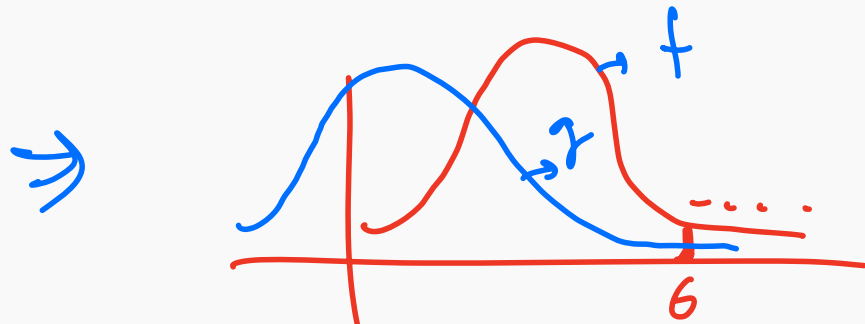
$\hat{\theta}_n^{\text{IS}}$ is **unbiased** for θ , for any valid choice of g – exactly like crude MC (which is the special case $g = f$, giving all weights $w_i \equiv 1$).

$$\begin{aligned} \theta &= \int f(x) h(x) dx \\ &= \int \frac{f(x) h(x)}{g(x)} g(x) dx \\ &= \int \frac{f(x) h(x)}{g(x)} \frac{1}{n} \sum_{i=1}^n g(X_i) dX_i \end{aligned}$$

$$\begin{aligned} E[\hat{\theta}_n^{\text{IS}}] &= \theta \\ E\left[\frac{1}{n} \sum_{i=1}^n w_i h(X_i)\right] &= \frac{1}{n} \sum_{i=1}^n E[w_i h(X_i)] \\ &= \frac{1}{n} \sum_{i=1}^n E\left[\frac{f(X_i)}{g(X_i)} h(X_i) g(X_i)\right] \\ &= \frac{1}{n} \sum_{i=1}^n \int \frac{f(x) h(x)}{g(x)} g(x) dx \\ &= \theta \end{aligned}$$

How to Choose the Proposal g ?

- The variance of the importance sampling estimator is $\text{Var}_g[w(X)h(X)]/n$, where $w(x) = f(x)/g(x)$. A poor g can **inflate** variance – even to infinity if g has lighter tails than $f \cdot h$.
- Put more mass of g where $|h(x)|f(x)$ is large – e.g. shift g toward the tail for rare-event problems, exactly as we shifted the envelope for truncated-Normal rejection sampling in Module 1.
- Heavy tails matter:** if g 's tail is lighter than f 's, the weight $w(x) = f(x)/g(x)$ can blow up, giving a handful of enormous weights that dominate the estimator and destroy its variance – a common and dangerous pitfall.
- Rule of thumb:** prefer g with tails at least as heavy as f ; check that $\text{Var}_g[w(X)h(X)] < \infty$ before trusting the estimator.



⇒

$$\begin{aligned} \theta_n^{\text{IS}} &= \frac{1}{n} \sum_{i=1}^n w(x_i) h(x_i) \\ \text{Var}(\theta_n^{\text{IS}}) &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(w(x_i) h(x_i)) \\ &= \frac{1}{n} \text{Var}(w(x) h(x)) \end{aligned}$$

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- The variance of the importance sampling estimator is $\text{Var}_g[w(X)h(X)]/n$, where $w(x) = f(x)/g(x)$. A poor g can **inflate** variance – even to infinity if g has lighter tails than $f \cdot h$.
- Put more mass of g where $|h(x)|f(x)$ is large – e.g. shift g toward the tail for rare-event problems, exactly as we shifted the envelope for truncated-Normal rejection sampling in Module 1.
- **Heavy tails matter:** if g 's tail is lighter than f 's, the weight $w(x) = f(x)/g(x)$ can blow up, giving a handful of enormous weights that dominate the estimator and destroy its variance – a common and dangerous pitfall.
- **Rule of thumb:** prefer g with tails at least as heavy as f ; check that $\text{Var}_g[w(X)h(X)] < \infty$ before trusting the estimator.

Practical diagnostic

Look at the distribution of the weights $w_1, \dots, w_n$ themselves – if a tiny number of draws carry almost all the weight, the estimator is unreliable no matter how large n is. (We formalize this with **effective sample size** later in the module.)

Rare Event via Importance Sampling

Recall: $\theta = P(X > 6)$, $X \sim N(0, 1)$, $\theta \approx 9.87 \times 10^{-10}$; crude MC needed billions of draws.

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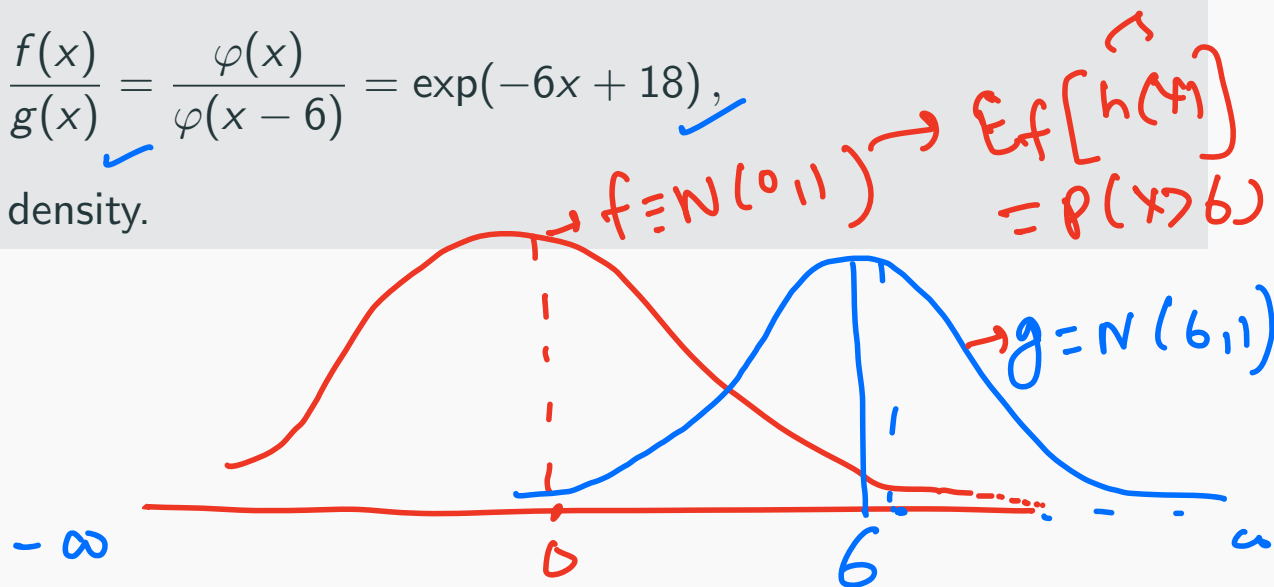
Importance sampling fix: shift the proposal into the tail

Take $g(x) = N(6, 1)$ density (i.e. shift the mean to the boundary of the rare event). Then

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where φ is the standard Normal density.

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Algorithm

1. Draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(6, 1)$ – now roughly *half* the draws exceed 6!

2. $\hat{\theta}_n^{\text{IS}} = \frac{1}{n} \sum_i \exp(-6X_i + 18) \mathbf{1}\{X_i > 6\}.$

Handwritten:

$$\downarrow$$
$$\text{CML} \rightarrow \frac{1}{n} \sum_{i=1}^n \mathbf{I}(X_i > 6) \rightarrow \frac{1}{n} \sum_{i=1}^n \underbrace{\exp(-6X_i + 18)}_{w(x_i)} \underbrace{\mathbf{1}\{X_i > 6\}}_{h(x_i)} \Rightarrow \frac{1}{n} \sum_{i=1}^n w(x_i) h(x_i)$$

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With just $n = 10,000$ draws from the shifted proposal, the estimate is accurate to several significant figures – a dramatic improvement over crude MC's complete failure.

When f Is Known Only Up to a Constant

- In Bayesian problems (recall the genomics/AI evidence example), the target is often $f(x) = \tilde{f}(x)/c$ where $\tilde{f}$ is computable but the normalizing constant $c = \int \tilde{f}$ is **unknown**.

$$\begin{aligned} &\Rightarrow f \Rightarrow \text{density} \\ &\Rightarrow E_f[h(x)] = \int h(x) f(x) dx \\ &\tilde{f}(x) > 0 \Rightarrow \int \tilde{f}(x) dx \neq 1 \\ &\int \tilde{f}(x) dx = c < \infty \\ &\Rightarrow f = \frac{\tilde{f}(x)}{c} \rightarrow f: \text{ proper pdf } \end{aligned}$$

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Algorithm: Weighted (Self-Normalized) Importance Sampling

1. Generate $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} g$.
2. Compute $\tilde{w}_i = \tilde{f}(X_i)/g(X_i)$.
3. Return

$$\hat{\theta}_n^{\text{WIS}} = \frac{\sum_{i=1}^n \tilde{w}_i h(X_i)}{\sum_{i=1}^n \tilde{w}_i}.$$

$$\rightarrow \theta = \mathbb{E}f(h(x)) = \int h(x) f(x) dx$$

Sketch: Consistency of the Weighted Estimator

- Write the estimator as a ratio of sample means:

$$\hat{\theta}_n^{\text{WIS}} = \frac{\frac{1}{n} \sum_{i=1}^n \tilde{w}_i h(X_i)}{\frac{1}{n} \sum_{i=1}^n \tilde{w}_i} =: \frac{A_n}{B_n}, \quad X_i \stackrel{\text{iid}}{\sim} g.$$

$$\hat{\theta}_n^{\text{WIS}} = \frac{\frac{1}{n} \sum_{i=1}^n \tilde{w}_i(X_i) h(X_i)}{\frac{1}{n} \sum_{i=1}^n \tilde{w}_i(X_i)} \rightarrow \frac{A_n}{B_n} \Rightarrow$$

$$A_n = \frac{1}{n} \sum_{i=1}^n \tilde{w}_i(X_i) h(X_i) \xrightarrow{P} \mathbb{E}_g[\omega(X) h(X)] \rightarrow \sigma \text{ as } n \rightarrow \infty$$

$\tilde{w}_i \xrightarrow{\text{iid}} \omega$

$$\Rightarrow \text{Var}(\tilde{w}_i(X_i) h(X_i)) = \text{Var}(\omega(X) h(X)) = \sigma^2 < \infty$$

$$B_n = \frac{1}{n} \sum_{i=1}^n \tilde{w}_i(X_i) \xrightarrow{P} \mathbb{E}_g(\omega(X)), \quad n \rightarrow \infty$$

$$\Rightarrow \begin{matrix} X_n \xrightarrow{P} X \\ Y_n \xrightarrow{P} Y \end{matrix} \Rightarrow \frac{X_n}{Y_n} \xrightarrow{P} \frac{X}{Y}$$

$$\Rightarrow \begin{matrix} A_n \rightarrow A \\ B_n \rightarrow B \end{matrix} \Rightarrow \frac{A_n}{B_n} \rightarrow \frac{A}{B}$$

$$\hat{w}(x) = \frac{\hat{f}(x)}{\hat{g}(x)} = \frac{c f(x)}{g(x)}$$

$$f(x) = \frac{\tilde{f}(x)}{c} \Rightarrow \hat{f}(x) = c f(x)$$

$$\tilde{f}(x)$$

$$\Rightarrow \hat{\theta}_n^{wS} = \frac{\frac{1}{n} \sum_{i=1}^n \tilde{w}(x_i) h(x_i)}{\frac{1}{n} \sum_{i=1}^n \tilde{w}(x_i)} \xrightarrow{P} \frac{Eg(\tilde{w}(X) h(X))}{Eg(\tilde{w}(X))}$$

$$= \frac{\int \cancel{c} \frac{f(x)}{\cancel{g(x)}} \cdot h(x) \cancel{g(x)} dx}{\int \cancel{c} \frac{f(x)}{\cancel{g(x)}} \cdot \cancel{g(x)} dx} = \frac{\int f(x) h(x) dx}{\int f(x) dx} = \frac{E_f(h(x))}{1} = E_f(h(x))$$

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- Therefore $\hat{\theta}_n^{\text{WIS}} = A_n/B_n \rightarrow (c\theta)/c = \theta$ (consistency).

Sketch: (Non-)Unbiasedness of the Weighted Estimator

- Even though $E[A_n] = c\theta$ and $E[B_n] = c$, for large n ; and in general

$$E\left[\frac{A_n}{B_n}\right] \neq \frac{E[A_n]}{E[B_n]}.$$



$$E\left(\frac{X}{Y}\right) \neq \frac{E(X)}{E(Y)}$$

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- The good news: the bias usually decays like with large n .